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UKUDLA Data Science Certificate

Reproducible Environments using `renv` & Docker

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The reproducibility question

Can another learner

- run the analysis scripts?
- obtain the required packages?
- reproduce the same results?
- recreate the analysis next year?

Environments can differ

Learner A	Learner B
R 4.3.3	R 4.5.0
dplyr 1.1.4	dplyr 1.2.x
Ubuntu	Windows

Sharing code is necessary, but code alone does not describe the computational conditions under which it ran. “Works for me” is not yet a reproducibility strategy.

A computational environment has several layers

Project code, configuration and data
R packages and package versions
R version and publishing tools
System libraries and operating system
Computer infrastructure

Isolation
Keep project dependencies separate.

Portability
Move work between computers.

Reproducibility
Recreate recorded conditions.

What does `renv` manage?

`renv` provides

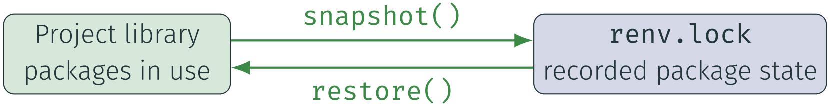
- a project-local R package library;
- isolation from other projects;
- a record of package versions and sources; and
- tools to restore that state.

`renv` does not manage

- the complete operating system;
- all system libraries;
- hardware or external services; or
- the research workflow itself.

`renv.lock` is a machine-readable description of the project's R package environment and belongs in version control.

The `renv` workflow



- `snapshot()` updates the lockfile from the project library.
- `restore()` recreates the project library from the lockfile.
- Review lockfile changes before committing them.
- A shared project can be initialized once and restored by collaborators.

What does a container add?

A container packages

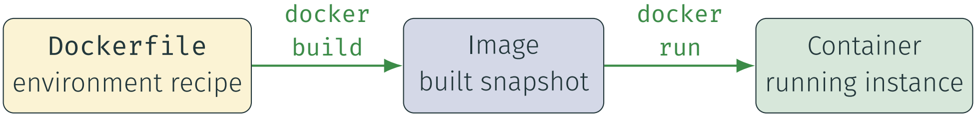
- an operating-system user space;
- system libraries and tools;
- R and publishing tools;
- project dependencies; and
- a default working context.

A container shares

- the host computer's kernel;
- host hardware and resources;
- explicitly mounted files; and
- configured network access.

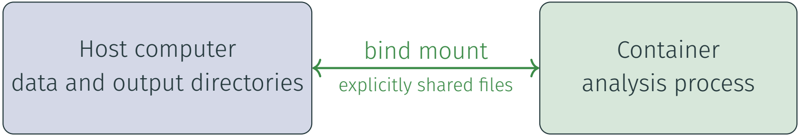
Containers provide stronger environment isolation than a project package library, but are lighter than a complete virtual machine.

Dockerfile, image and container



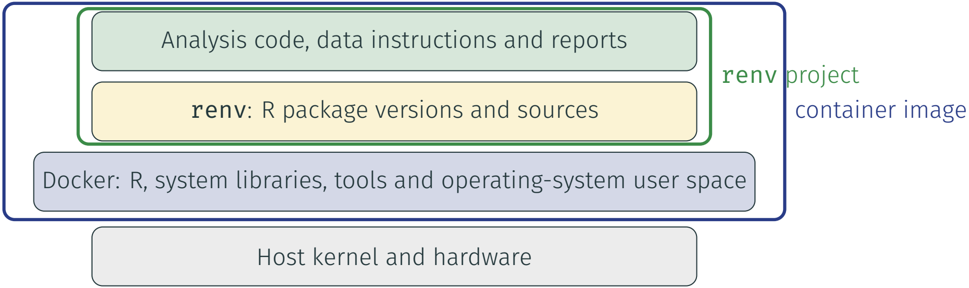
- The Dockerfile describes how to construct an image.
- The image is an immutable snapshot that can be used repeatedly.
- Each container is a running instance created from that image.
- Rebuild after changing the Dockerfile or copied source code.

Containers are disposable; results must persist



- A container's writable layer normally disappears with the container.
- Bind mounts expose selected host directories inside the container.
- Inputs can be read and outputs written back to the host.
- Mount only what the analysis needs.

Two tools manage different layers



`renv` records the R package layer. Docker records a broader execution environment. Together they provide complementary reproducibility information.

Run one analysis reproducibly across contexts

Method	Environment	Typical use
Local batch	Host R with restored **`renv`** library	Run the complete workflow
Local interactive	RStudio or terminal R with **`renv`**	Develop and inspect
Docker batch	Image in a temporary container	Run a consistent environment
Docker interactive	Shell or R inside the container	Inspect the container context

Git repository → Execution environment → Analysis workflow → Data, figures, report

Key message: record every relevant environment layer

Environment tools can help recreate

- package versions;
- language and system tools;
- execution commands; and
- a consistent working context.

They cannot guarantee

- correct data or code;
- valid scientific assumptions;
- permanent external sources; or
- responsible interpretation.

`renv` records R packages; Docker records a broader environment. Together with Git, documented data, and a reproducible workflow, they make an analysis easier to recreate and inspect.

Detailed setup, repository, workflow, and troubleshooting instructions are provided in the learning-module guides.

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